Handy Helper Umbrella

8-Panel Field Reference for HVAC & Electrical Trades

1 Safety First

LOTO, PPE & Wet-Work Precautions

Discussion Aid, Not a Work Procedure[1]

Reinforce discipline, work ethic and communication. Does not replace official paperwork, training, authorization or equipment-specific procedures. Keep job records private.

Use the employer's applicable energy-control procedure. Discuss all sources, stored energy, reaccumulation, verification, roles and group/shift continuity. A disconnect or meter reading alone does not establish safety.

Refer required energized electrical testing to qualified, authorized personnel. Follow approved inspection, personnel-clearance, device-removal, notification and restart procedures; this aid does not authorize removal of another person's lock.

Wet-Work Precautions[1]

Do not wash energized equipment or direct liquid into electricals. Follow exact equipment/coil restrictions. Covers, GFCIs and drying do not substitute for isolation. Electrical tools must suit the location.

Task-Specific Protection[11]

Select shock/arc, eye/face, chemical, hearing, foot and other protection from the assessed hazards. No universal glove class or clothing list establishes a complete protective system.

1. OSHA hazardous energy: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.147
Electrical work: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.333
11. PPE: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.335

2 Refrigeration Cycle

The 4 Stages of the Cycle

The 4 Stages[2,3]

1. Compressor — raises pressure & temperature of vapor
2. Condenser — rejects heat, vapor becomes liquid
3. Metering Device — drops pressure, liquid becomes mixed phase
4. Evaporator — absorbs heat, refrigerant becomes vapor

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COMPRESSOR
^ v
Low-P High-P
vapor vapor
^ v
EVAP CONDENSER
^ v
<-- METERING DEVICE -->
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R410A PT Conversions[10]

350-450 PSIG corresponds to 106.8-125.4°F saturated
100-140 PSIG corresponds to 31.4-48.9°F saturated
Conversions only, not operating or charging targets.

Metering Devices: TXV (best efficiency) | Fixed piston (charge-sensitive) | Cap tube (small systems) | EEV (electronic, variable)

2. HVAC School: https://www.hvacrschool.com/
3. Anthony's Cool School HVAC Training: https://hvac.anthonyscoolschool.com/
10. Honeywell AZ-20 PT chart (Feb 2019): https://prod-edam.honeywell.com/content/dam/honeywell-edam/pmt/oneam/en-us/refrigerants/documents/pmt-am-genetron-aZ20-pressure-temp-chart-tech-tool1.pdf

3 Superheat & Subcooling

Charge & Performance Diagnostics

Superheat = Suction Line Temp − Sat. Suction Temp[4]

Use existing authorized readings with paired pressure/temperature locations and identified refrigerant.

Consideration	Required context
Target	Equipment method and metering device
Conditions	Operating mode, airflow, load, stability

Subcooling = Sat. Liquid Temp − Liquid Line Temp[4]

Compare with the model-specific target; no universal normal range is supplied.

Consideration	Required context
PT selection	Correct refrigerant, units and phase
Interpretation	Not a one-reading charge diagnosis

Calculate from authorized data, then use the manufacturer's diagnostic method. Pressure patterns alone do not authorize charge adjustment. Confirm applicable Section 608 qualifications before refrigerant work.

4. HVAC School - Superheat & Subcooling: https://www.hvacrschool.com/
EPA certification: https://www.epa.gov/section608/section-608-technician-certification-requirements

4 Thermostat Wiring

Low-Voltage Terminal Reference

Terminal Reference[5]

Term	Color	Function
R/RH/RC	Red	24V power (heat/cool)
C	Blue	Common return
Y/Y1	Yellow	Cooling -> contactor
G	Green	Fan / blower
W/W1	White	Heating
O	Orange	Rev. valve - cool (most)
B	Dk Blue	Rev. valve - heat (Ruud/Rheem)
W2	White	Aux / 2nd-stage heat

Common Failures

No 24V at stat: Check transformer, fuse, float switch
Stat blank: Check R-C for 24V, C wire connection
Fan won't stop: G wire shorted, relay stuck
No cooling: Y not reaching contactor, float/pressure switch
Rev. valve wrong: Check O/B — energize-in-cool vs heat

Float switch: Breaking R kills stat + defrost board. Breaking Y kills cooling only.

5. HVAC School - Low Voltage Wiring: https://www.hvacrschool.com/videos/understanding-low-voltage-wiring-for-ac-heat-pumps-3d/

5 Contactor & Capacitor

Motor Circuit Testing

Contactor[6]

Coil: 24V from stat Y -> safety switches -> contactor coil
Contacts: 240V through contacts to compressor + condenser fan
Concepts only; use the exact schematic. A click or open contactor does not prove isolation. Refer required energized testing to qualified, authorized personnel.

Capacitor Terminals

Terminal	Connects To
C (Common)	Run winding + power
HERM	Compressor
FAN	Condenser fan motor

WARNING: Capacitors hold charge even when power is off!
Use the manufacturer's stored-energy discharge, waiting and verification procedure with approved rated tools. No universal resistor value or wait time; do not improvise terminal shorting.[1]

Testing

Inspect/test only under the applicable isolation and stored-energy procedure. Compare MFD with component marking and manufacturer tolerance. Continuity alone does not certify insulation or safe operation.

6. HVAC School: https://www.hvacrschool.com/
Stored energy: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.333

6 Heat Pump & Defrost

Sequence & Troubleshooting

Heating Mode Sequence[7]

1. Stat calls heat (W/Y energizes)
2. Reversing valve positions for heating (de-energized on most brands)
3. Compressor starts — outdoor coil = evaporator (absorbs heat)
4. Indoor coil = condenser (rejects heat)
5. Supplemental heat (W2) engages if needed

Defrost Cycle

Triggers when ice builds on outdoor coil (typically <40F outdoor). Defrost board reverses to cooling mode, outdoor fan stops, heat strips energize. Duration: 5-15 min.

Problem	Check
Never defrosts	Defrost sensor/board, reversing valve
Defrosts too often	Sensor location/accuracy
Won't exit defrost	Sensor stuck, board relay
Ice on coil	Low charge, low airflow, defrost fail

Rev. valve: Most brands energize O in cooling. Ruud/Rheem energize B in heating.

7. HVAC School: https://www.hvacrschool.com/
Anthony's Cool School: https://hvac.anthonyscoolschool.com/

7 Electrical Formulas

Ohm's Law, Power & Voltage Drop

Ohm's Law[8]

V = I x R I = V / R R = V / I

Power (Single Phase)

P = V x I P = I² x R P = V² / R

Three-Phase Power

P = V x I x 1.732 x PF (PF typically 0.8-0.95)

Voltage Drop (Copper)[8]

VD = 2 x 12.9 x I x D / CM

K=12.9 (copper) | I=amps | D=one-way feet | CM=circular mils

AWG	Ampacity (60C)	CM
14	15A	4,107
12	20A	6,530
10	30A	10,380
8	40A	16,510
6	55A	26,240

Motor FLA estimate: 240V 1-phase ~= HP x 4.5 | 3-phase ~= HP x 2.5

Always verify against NEC Article 310 and local code.[8]

8. NFPA 70 - National Electrical Code: https://www.nfpa.org/codes-and-standards/detail?code=70

8 Troubleshooting Flow

Decision Trees & PM Checklist

No Cooling[9]

Stat on COOL, below room temp?

+-- No -> Check stat, batteries, display

+-- Yes -> Outdoor unit running?

+-- No -> Breaker -> disconnect -> contactor -> cap -> compressor

+-- Yes -> Compare authorized readings, airflow, load and mode

+-- Confirm refrigerant, paired locations and stable conditions

+-- Follow the manufacturer's diagnostic method

+-- Pressure patterns alone do not prove a fault or justify charging

Short Cycling

Check: stat location (drafts), filter, coil freeze-up, low charge, oversized equip, pressure switch

Coil Cleaning Discussion[12]

Use exact coil/equipment instructions for chemicals, pressure, distance, direction and tools. Do not assume pressure washers or chemicals are permitted. Follow approved isolation and return-to-service procedures.

9. Anthony's Cool School: https://hvac.anthonyscoolschool.com/

OSHA Safety: https://www.osha.gov/control-hazardous-energy

12. TMAL-specific cleaning restrictions: https://docs.johnsoncontrols.com/ductedsystems/api/khub/documents/TUg78wHltRmyi4kpDTod4w/content